Dr. Vittorio Saggiomo

Associate Professor (UHD) in BioNanoTechnology
Wageningen University & Research, The Netherlands

CONTACT INFORMATION

Field	Details
Full Name	Dr. Vittorio Saggiomo
Current Title	Associate Professor (UHD) in BioNanoTechnology
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RESEARCH INTERESTS

- Open hardware and open-source scientific instrumentation
- Laboratory automation and robotics (liquid handling, syringe pumps, closed-loop systems)
- Microfluidics and cleanroom-free microfabrication (ESCARGOT)
- Frugal point-of-care diagnostics and field-deployable analytical tools
- Optical materials, nanotechnology, and dichroic nanocomposites
- Environmental sensing (microplastics, eDNA, particle detection)
- Democratization of science through accessible, reproducible instrumentation

EDUCATION

Ph.D. in Chemistry (Summa cum Laude) — 2010

Christian-Albrechts-Universität zu Kiel, Kiel, Germany

- **Thesis Focus:** Dynamic Combinatorial Chemistry, macrocycles, and supramolecular host-guest chemistry.
- **Award:** Otto-Diels Prize (2010) — awarded for the best PhD thesis at the Otto-Diels Institute, CAU Kiel.

M.Sc. in Chemistry — 2007

Università degli Studi di Napoli Federico II, Naples, Italy

- **Focus:** Organic Chemistry and Molecular Systems.

PROFESSIONAL & ACADEMIC POSITIONS

Associate Professor (Open Hardware and Sensing) — 2023 – Present

Wageningen University and Research (WUR) | Wageningen, The Netherlands *(Assistant Professor 2017 – 2023)*

Postdoctoral Research Fellow — 2013 – 2016

Wageningen University and Research (WUR) | Wageningen, The Netherlands

Postdoctoral Research Fellow (Systems Chemistry) — 2010 – 2013

University of Groningen | Groningen, The Netherlands

RESEARCH LEADERSHIP & GROUP

- Established and currently lead a state-of-the-art **Open Hardware microfabrication and sensing laboratory**, with a team of **12 PhD candidates (6 completed, 6 current)** and **3 postdocs**.
- Produced **2 granted international patents** in low-cost microfluidic manufacturing.
- Part of the laboratory is dedicated to sensing, translating complex physical and optical concepts into validated, automated scientific instruments deployed globally.
- **Primary interface** across international consortia, coordinating technical deliverables, system constraints, and data goals among engineers, physicists, and environmental scientists.

PROJECT MANAGEMENT & GRANTS

- Directed **multi-disciplinary research consortia across 17+ competitive grants**.
- Secured over **€10M** in total portfolio funding (€2.4M+ direct to WUR) to engineer automated, open-source instrumentation platforms.
- Demonstrated track record of securing competitive national, European, and institutional funding dedicated to open science ecosystems, open hardware instrumentation, and educational democratization.

GRANTS & FUNDING PORTFOLIO

> ***Total Portfolio:*** *Exceeding* ***€10 Million*** *(over €2.4 Million direct to WUR across 17+ granted applications).*

Major Open Science, Infrastructure & Consortium Grants

Grant	Year	Role	Budget
Open Science Infrastructure Large Grant	2025 (Kicking off Summer 2026)	Co-Applicant / Core Group Developer	–
OSH-NED (Open Science Hardware Nederland)	–	Developing architectural foundation and open resource network for national Open Science Hardware framework	–
Marie Skłodowska-Curie Doctoral Network (MSCA-DN)	2022 – 2023	Co-PI / Partner (with One Planet)	€270k (WUR)
4TU Green Sensors Grant	2022	Co-PI (Consortium across Dutch Technical Universities)	€3.0M Total (€120k WUR)
ERRAZE + De Nieuwe Waarde Grant	2022	Co-PI (with Wageningen Entomology)	€290k
Open Science Fund Grant	2023	Principal Investigator (with Dr. Namita Sharma)	€50k
TKI LSH Health / ID4Corona Grant	2020	Co-PI (with TNO Microbiology)	€196k Total (€29k WUR)

Innovation & High-Risk Exploratory Grants (NWO XS / Proof-of-Concept)

Grant	Year	Role	Budget	Focus
NWO Take-Off Grant (Phase 1)	2020	Principal Investigator	€40k	Commercial feasibility and tech transfer of open microfluidic molding technologies

Grant	Year	Role	Budget	Focus
NWO XS Grants (High-Risk/Exploratory)	2019 – 2021	Principal Investigator / Co-PI	Multiple €50k awards	Blue-sky projects including open-hardware amyloids analysis, flow-NMR automation, low-cost microplastics particle sorting arrays
4TU Plantenna & NWA Plantenna Consortia	2018	Co-Investigator (WUR lead team)	€4.3M Total (€720k WUR)	Ultra-low-power, plant-integrated climate and environmental sensor platforms
MSCA-RISE Grant (IPANEMA)	2019	Co-Investigator	€1.35M Total (€133k WUR)	International staff exchange network advancing microfluidic monitoring and frugal diagnostic platforms

Open Science Education & Internal Seed Grants

Grant	Year	Role	Budget	Focus
Open Science Education Pilot Grant	2026	Principal Investigator	–	Developing interactive, open-source hardware readers
The Open Science Education Grant	2026	Principal Investigator	–	Reusing Old Hardware with Python (educational methodology to combat electronic waste)
WUR Wildcard Grant & VLAG PhD/Postdoc Grants	2017 – 2022	Principal Investigator / Co-PI	€50k – €260k	Funding cross-departmental research bridging BioNanoTechnology with Food Quality & Design and Biobased Process Engineering
Early Career Fellowships (Rubicon & Marie Curie IEF)	2011 – 2012	Principal Investigator	NWO Rubicon: €110k & EU Marie Curie: €144k	Supramolecular chemistry and dynamic combinatorial library design

RESEARCH & OPEN HARDWARE INNOVATIONS

Driven by the core mission of **democratizing science**, research focuses on removing economic and logistical barriers to high-precision research by replacing expensive, proprietary “black-box” laboratory equipment with open-source, affordable, and modifiable alternatives.

1. Transforming 3D Printers into Open Laboratory Robotics

- **Repurposing Consumer Motion Platforms:** Pioneered G-code firmware modification of consumer 3D printers, converting sub-€300 motion platforms into high-precision laboratory robots (syringe pumps, slide stainers, scanning stages) that reduce equipment acquisition costs by **>80%**.
- **Closed-Loop Lab Automation:** Developed open workflows transforming desktop 3D printers into automated liquid handlers, fraction collectors, automated slide stainers (**HistoEnder**), and self-optimizing closed-loop chemical reactors.
- **Precision Fluidic Actuation:** Engineered open-source conversion kits turning standard 3D printers into multi-channel syringe pump arrays, enabling micro-metered fluidic dispense system integration with computer vision at a fraction of commercial costs.

2. Frugal Microfluidics & Advanced Fabrication

- **Cleanroom-Free Prototyping:** Shifted microfluidic chip fabrication out of traditional, high-cost cleanrooms by engineering SLA/FDM 3D-printed molds.
- **PDMS Molding Techniques:** Invented the **ESCARGOT** scaffold-removal microfabrication technique, enabling cleanroom-free PDMS microfluidic chip fabrication down to **25 µm** resolution. Generating **>260** citations and **>110,000** views on open-access tutorial repositories.
- **Sustainable Bio-Resins:** Innovated eco-friendly digital fabrication processes by formulating renewable SLA 3D printing resins derived from microalgae biomass.

3. Open Microscopy & Field-Ready Optical Instrumentation

- **EnderScope:** Co-developed an automated, 3D printer-based scanning microscope engineered for automated microplastic, eDNA, and particle detection. Open alternative to costly commercial scanning stages.
- **ESPressoscope & Portable Optics:** Engineered the **ESPressoscope**, an ultra-low-cost (~€10) ESP32-based Wi-Fi scanning microscope, enabling real-time wireless micro-imaging in resource-limited field settings. Reducing hardware costs by **>95%** compared to commercial alternatives.
- **Automated & In-Situ Imaging:** Integrated low-cost image sensor arrays with custom microcontrollers and computer-vision pipelines to monitor biological and chemical phenomena in real time.

4. Optical Materials, Nanotechnology & Dichroic Nanocomposites

- **Lycurgus Cup Optical Replicas:** Pioneered synthesis and 3D printing of gold and silver nanoparticle-embedded polymer nanocomposites exhibiting dichroic optical properties (changing color in transmission vs. reflection, inspired by the Roman Lycurgus Cup).
- **Custom Optical Elements:** Investigated wavelength-dependent light scattering and absorption properties of embedded nanomaterials for digital fabrication of custom optical filters and sensing elements.

5. Point-of-Care Diagnostics & Frugal Analytical Tools

- **T-Cup & Coffee Capsule Testing:** Invented the **T-Cup** (*Ex coronaspresso*) point-of-care diagnostic device using upcycled coffee pods and 3D printing. Achieving instrument-free LAMP detection of viral RNA (SARS-CoV-2) and securing national TV/press coverage across **6+** major media outlets.
- **On-Site Field Analysis:** Engineered simplified DNA extraction protocols combined with handheld heating arrays for rapid, on-site biological detection without standard lab infrastructure.

6. Tropical Disease Diagnostics & Field Deployment

- **Schistosomiasis Detection:** Deployed **25 µm** staircase microfluidic sorting chips and MatchboxScopes in pilot field studies for detecting parasite eggs (Schistosoma) in urine in collaboration with LUMC (Leiden) and field teams in Lagos, Nigeria.

7. Bioinspired Materials & Live Publishing Workflows

- **Bioinspired Probabilistic Fastener:** Engineered 3D-printed mushroom-inspired adhesive surfaces (Biointerphases, 2021) that received widespread international media coverage.
- **Live Publishing Innovation:** Early adopter of dynamic “live publishing” workflows, continuously updating published open hardware designs dynamically via GitHub and PubPeer repositories.

PUBLICATIONS

Bibliometric Summary

Metric	Value
Total Citations	**>1,700** (OpenAlex, October 2026)
h-Index	**23** (OpenAlex)
i10-Index	**40** (OpenAlex)
Indexed Works	136 (OpenAlex)

Representative Peer-Reviewed Publications

Open Hardware & Lab Automation

- Saggiomo, V. (2022). “A 3D Printer in the lab: Not only a toy.” *Advanced Science*, 9(22), 2201274.

- Saggiomo, V. (2022). “Open hardware is at the heart of device research.” *Device* (Cell Press), 1(1), 100002.
- Baas, S., & Saggiomo, V. (2021). “Ender3 3D printer kit transformed into open, programmable syringe pump set.” *HardwareX*, 10, e00219.
- Ponzetti, M., et al. (2022). “HistoEnder: a 3D printer-based histological slide autostainer.” *HardwareX*, 12, e00370.
- Del Rosario, M., et al. (2022). “The field guide to 3D printing in optical microscopy for life sciences.” *Advanced Biology*, 6(4), 2100994.

Microfluidics, Materials Science & Frugal Diagnostics

- Saggiomo, V., & Velders, A. H. (2015). “Simple 3D Printed Scaffold-Removal Method for the Fabrication of Intricate Microfluidic Devices.” *Advanced Science*, 2(9), 1500063. *(>260+ citations, >80k YouTube views)*
- Velders, A. H., et al. (2022). “T-Cup: A Cheap, Rapid, and Simple Home Device for Isothermal Nucleic Acid Amplification.” *Global Challenges*, 6(3), 2100078.
- Kool, L., et al. (2020). “Gold and silver dichroic nanocomposite in the quest for 3D printing the Lycurgus cup.” *Beilstein Journal of Nanotechnology*, 11, 16–23.
- Iuliano, A., et al. (2020). “Coupling 3D Printing and Novel Replica Molding for In House Fabrication of Skeletal Muscle Tissue Engineering Devices.” *Advanced Materials Technologies*, 5(9), 2000344. **(Best Paper of 2020)**
- Li, E., et al. (GitHub: matchboxscope). **ESPressoscope**: Efficient Swiss-Army knife for Photonics.
- Velders, A. H., et al. (2022). “Step-by-step: A microfluidic (PDMS) staircase device for size sorting microparticles down to 25um.” *ChemRxiv*.

PATENTS

Patent 1: Method of Manufacturing Microdevices for Lab-on-a-Chip Applications

- **Patent Numbers:** US Patent 11,052,388 | WO Patent 2021/251,816
- **Inventors:** W.W.M. Pim Pijnappel, Jessica C. de Greef, Vittorio Saggiomo
- **Assignees:** Erasmus MC, LUMC, Wageningen University
- **Summary:** Advanced replica molding method utilizing 3D-printed sacrificial and stretchable molds for fabricating elastomeric microfluidic arrays and tissue engineering chips (muscle-on-a-chip).

Patent 2: Method of Manufacturing a Microfluidic Device (ESCARGOT)

- **Patent Number:** NL/EP Patent 2016155760
- **Inventors:** Vittorio Saggiomo, Aldrik H. Velders
- **Summary:** Embedded Scaffold Removing Open Technology (**ESCARGOT**) for cleanroom-free, two-step microfluidic channel fabrication using 3D-printed ABS scaffold dissolution.

TEACHING, MENTORSHIP & PEDAGOGICAL HIGHLIGHTS

> *A recognized pioneer in **Open Science Education** and **“Maker” pedagogy**, dedicated to transforming students from passive instrument consumers into active, critical hardware builders.*

Pedagogical Vision & Teaching Philosophy

- **The Modern Polymath Toolset:** Believes modern life scientists and chemists must be comfortable with 3D design, digital fabrication, and microcontrollers alongside molecular science.
- **Open Protocols & Learning Through Failure:** Moved away from traditional “cookbook” lab guides. Provides open-ended goals where students are not penalized for failed experiments.
- **Interdisciplinary & Flexible Paths:** Teaches multidisciplinary cohorts (Molecular Life Sciences, Biotechnology, Food Sciences, Environmental Sciences, and Plant Sciences) using collaborative, problem-based challenges.

Primary Course Coordination & Curriculum Development

BioNanoTechnology: Sensors & Devices (BNT-51306)

- **Scope:** 6 ECTS course designed and created from scratch. Covers physical/chemical sensing principles, microfluidics, physical computing (Arduino/microcontrollers), and digital fabrication.

- **Student Ratings:** Consistently awarded outstanding evaluations (**4.4–4.9 / 5.0**).
- **Experiments@Home Box:** Developed and deployed the hybrid kit to 100+ remote students during COVID lockdowns, earning a feature in *Nature Reviews Chemistry*. Now integrated permanently as a hybrid learning module.

Open Hardware in the Lab: DIY using 3D printers, laser cutters, and CNC

- **VLAG Graduate School Instructor** (since 2024); course delivered again in **2026**.
- **Scope:** Specialized PhD course bridging CAD sketching, slicer optimization, and digital fabrication.
- **2026 Evaluation: 4.9 / 5.0** student rating.

Student Team Coaching & Competitions

Competition	Achievement
SensUs International Student Competition on Biosensors	2nd Place Globally for Analytical Sensitivity (inaugural 2022 entry)
iGEM (International Genetically Engineered Machine)	Co-mentored hardware, microfluidic interfaces, and sensor assembly
NWO Chemistry Student Competition	1st Place Victory in 2017

International Expert Training & Pedagogical Outreach

Role	Details
EMBO Practical Course Expert Trainer	April 2026: Selected as international trainer for “Hack your microscope” in Oeiras, Portugal
Educational Open Access Publications	Author of “Experiments@home” (*Nature Reviews Chemistry*, 2021) and world’s first open-download Electron Microscopy Coloring Book

Supervision & Team Leadership Track Record

- **PhD & Postdoc Supervision: 6 PhD candidates completed** and **6 currently in progress**, plus **3 Postdocs**. 100% of completed doctoral researchers defended successfully. Led 30+ MSc/BSc thesis projects to completion.
- **Student Mentorship Volume:** Directly supervised 30+ MSc and BSc thesis projects, and hosted international visiting scholars.
- **Training Volume:** Trained 200+ undergraduate and doctoral researchers.

EDITORIAL, NETWORK & COMMUNITY LEADERSHIP

Role	Organization
Chief Editor	The Journal of Open Hardware (Ubiquity Press / Open Journal Systems)
Advisory Board Member	Device (Cell Press)
Co-Developer & Board Member	OSHNED (Open Science Hardware Nederland)
Core Group Member	Open Science Community Wageningen (OSC-W)
Active Community Member	Gathering for Open Science Hardware (GOSH)
National Expert	Primary point of contact for the Open Hardware Community in the Netherlands
Social Media Editor (Past)	Supramolecular Chemistry (Taylor & Francis)

PEER REVIEWING & INSTITUTIONAL SERVICE

Journal Reviewing

- Regularly reviews for *Nature Reviews Chemistry*, *Chemical Communications*, *Analyst*, *HardwareX*, and *Journal of Materials Chemistry C*.

Institutional Responsibilities (Wageningen University)

- **Data Management Steward:** Overseeing FAIR data management guidelines for the BioNanoTechnology (BNT) group.

- **Tenure Track & Entry Committee (BAC):** Serving on hiring and promotion panels across Agrotechnology & Food Sciences (AFSG).
- **PhD Examination Committees:** Internal and external committee member for doctoral defenses across European universities.
- **AFSG Library Committee Member:** Advising on institutional open-access publishing strategies.
- **VLAG Visitation Writing Team (2021):** Committee member for the VLAG Graduate School self-evaluation report.
- **hDMT Consortium:** Active member of the human Organ and Disease Model Technology (hDMT) skeletal-muscle-on-a-chip group.
- **SAFE 4TU Committee:** Steering Committee for SAFE (Semiconductor Advances for Future Electronics and Sensors).

OPEN SCIENCE LEADERSHIP & ADVOCACY

- **Open Science Hardware Nederland (OSH-NED) Core Developer:** Co-applicant and technical lead for the national open science hardware network connecting Dutch universities and research institutes.
- **Open Science Community Wageningen (OSCW):** Core Group Member leading institutional initiatives toward open research practices.
- **National Whitepaper Co-Author:** Contributed to the Dutch national whitepaper "Creating an Open-source Hardware Ecosystem for Research and Sustainable Development".
- **Global Open Science Hardware (GOSH) Movement:** Active contributor to global open hardware initiatives, advocating for equitable access to scientific tools.

KEYNOTE & GUEST LECTURES

Event	Year	Details
TEDxWageningen	2021	"Open Science for Inclusive Science"
IUPAC CHAINS	2023	Invited talk on "Open Technologies: From Microfabrication to Automation"
Analytica Munich	2022	Keynote: "3D Printers as Scientific Instruments"
International Guest Workshops	–	Lectures and DIY science workshops at Loughborough University (UK), Uppsala University (SE), Nanyang Technological University (SG), BAM Berlin (DE)

HIGH-IMPACT MEDIA, PRESS & PUBLIC OUTREACH

National & International Television/Radio

- **EditieNL (RTL) & EenVandaag:** Featured television interviews discussing the T-Cup (CoronaEspresso) home diagnostic kit and student biosensing inventions.
- **TEDxWageningen:** Mainstage speaker delivering "Open Science for Inclusive Science".

Major Print & Broadcast Media Features

- NRC Handelsblad, Algemeen Dagblad (AD), De Gelderlander: Dedicated full-page spreads on the T-Cup coffee-capsule diagnostic project.
- **The Daily Mail (UK):** Feature article on the 3D-printed dichroic nanocomposite mimicking the ancient Roman Lycurgus Cup.
- **New Scientist & Metro (UK):** Global coverage of frugal diagnostics and handheld biosensors.

Scientific Trade & Specialist News Highlights

- **Nature Reviews Chemistry:** Featured highlight on the Experiments@home educational lab kits.
- **Chemical & Engineering News (C&EN):** Multiple expert commentary profiles, including being named one of the "**25 Chemists You Should Follow on Twitter**" and coverage of the CoronaEspresso diagnostic.

- **Chemistry World & C2W:** Extensive coverage of 3D-printed microfluidics, bioinspired probabilistic fasteners, and open-source lab automation.

Public Engagement & Community Events

- **Tardigrade Hunt (Wageningen Science Weekend, 2026):** Organised and ran the annual hands-on microscopy hunt for children and families — sampling, slide preparation and live imaging of tardigrades — as part of the national Science Weekend. The hunt also runs every year during Biodiversity Day, and open-hardware demonstrations have featured at Maker Faire / Maker Days.

AWARDS & HONORS

- **Otto-Diels Prize (2010)** — Best PhD thesis at the Otto-Diels Institute, CAU Kiel.
- **Best Paper of 2020** — Advanced Materials Technologies (Iuliano et al.).
- **1st Place, NWO Chemistry Student Competition (2017)** — team coaching.
- **2nd Place Globally, SensUs International Biosensor Competition (2022)** — team coaching.
- Named one of the "**25 Chemists You Should Follow on Twitter**" — Chemical & Engineering News.

DIGITAL OUTREACH METRICS

- **Bluesky:** >1,200 followers — bsky.app/profile/vsaggiomo.bsky.social
- **YouTube:** >2,300 subscribers.
- **Tutorial Views:** >110,000 views on open-source fabrication tutorials.
- **Mission:** Democratizing hardware designs worldwide.

TECHNICAL SKILLS & COMPETENCIES

Category	Skills
Digital Fabrication & Microfabrication	FDM, SLA, DLP 3D printing; CAD design & slicer optimization; cleanroom-free microfluidic chip design; ESCARGOT; elastomeric PDMS molding; microparticle sorting arrays
Physical Computing & Lab Automation	Arduino, ESP32, Raspberry Pi; C++ / C# / MicroPython; 3D printer firmware hacking (Marlin G-code repurposing)
Chemical Synthesis & Nanotechnology	Supramolecular chemistry; dynamic combinatorial libraries; metallic nanoparticle synthesis; renewable SLA bio-resins
Frugal Point-of-Care Diagnostics	Isothermal nucleic acid amplification (LAMP); phase-change material thermal regulation (T-Cup); portable Wi-Fi micro-imaging (ESPressoscope)

LANGUAGES

- English (fluent)
- Italian (native)
- Spanish (working knowledge)

> *"Democratizing science through open hardware and accessible instrumentation."*

Document Information

- **Prepared:** 2026
- **Last updated:** October 2026 — figures reconciled to OpenAlex and confirmed supervision/funding records; added the 2026 VLAG "Open Hardware in the Lab" course (4.9/5) and the Science Weekend *Tardigrade Hunt*; removed Dutch, Twitter and the references section; added Bluesky.
- **Format:** Academic Curriculum Vitae (Markdown)